TOR INDEPENDENCE IS LOCAL ON THE FIBRE PRODUCT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Localisation of Tor modules of two algebras over a ring. Lemma \ref{lemma-tor-independent-flat}, cited at proof lines 15122-15125 and 15138-15140. Identifies the localised Tor modules appearing in the three conditions.
- Vanishing of a module can be checked after localising at primes. Algebra, Lemma \ref{algebra-lemma-characterize-zero-local}, cited at proof lines 15129-15130. Turns the local vanishing statement into global Tor independence.

Source reference. Original lemma 08HX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10795–10795. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10794–10794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let A, B be R-algebras. The following are equivalent*

(1) *A and B are Tor independent over R,*

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(2) *for every pair of primes* $\mathfrak{p} \subset A$ *and* $\mathfrak{q} \subset B$ *lying over the same prime* $\mathfrak{r} \subset R$ *the rings* $A_\mathfrak{p}$ *and* $B_\mathfrak{q}$ *are Tor independent over* $R_\mathfrak{r}$*, and*
(3) *For every prime* $\mathfrak{s}$ *of* $A \otimes_R B$ *the module*

$$Tor_i^R(A, B)_\mathfrak{s} = Tor_i^{R_\mathfrak{r}}(A_\mathfrak{p}, B_\mathfrak{q})_\mathfrak{s}$$

(where $\mathfrak{p} = A \cap \mathfrak{s}$*,* $\mathfrak{q} = B \cap \mathfrak{s}$ *and* $\mathfrak{r} = R \cap \mathfrak{s}$*) is zero.*

Proof. Let $\mathfrak{s}$ be a prime of $A \otimes_R B$ as in (3). The equality

$$\mathrm{Tor}_i^R(A, B)_\mathfrak{s} = \mathrm{Tor}_i^{R_\mathfrak{r}}(A_\mathfrak{p}, B_\mathfrak{q})_\mathfrak{s}$$

where $\mathfrak{p} = A \cap \mathfrak{s}$, $\mathfrak{q} = B \cap \mathfrak{s}$ and $\mathfrak{r} = R \cap \mathfrak{s}$ follows from Lemma 08HW. Hence (2) implies (3). Since we can test the vanishing of modules by localizing at primes (Algebra, Lemma 00HN) we conclude that (3) implies (1). For (1) $\Rightarrow$ (2) we use that

$$\mathrm{Tor}_i^{R_\mathfrak{r}}(A_\mathfrak{p}, B_\mathfrak{q}) = \mathrm{Tor}_i^R(A, B) \otimes_{(A \otimes_R B)} (A_\mathfrak{p} \otimes_{R_\mathfrak{r}} B_\mathfrak{q})$$

again by Lemma 08HW. □